Transformations

Covering

- Transformations

Transformations

Transformation boil down to 4 key operations: Reflection, Rotation, Translation, and finally Enlargement.

Reflection, Rotation and Translation are isometric transformations, they do not change the shape. It makes equal angles and congruent shapes to the original figure.
Where as Enlargement is a Non-isometric transformation, it makes a larger or smaller shape. It makes equal angles but similar shapes to the original figure.

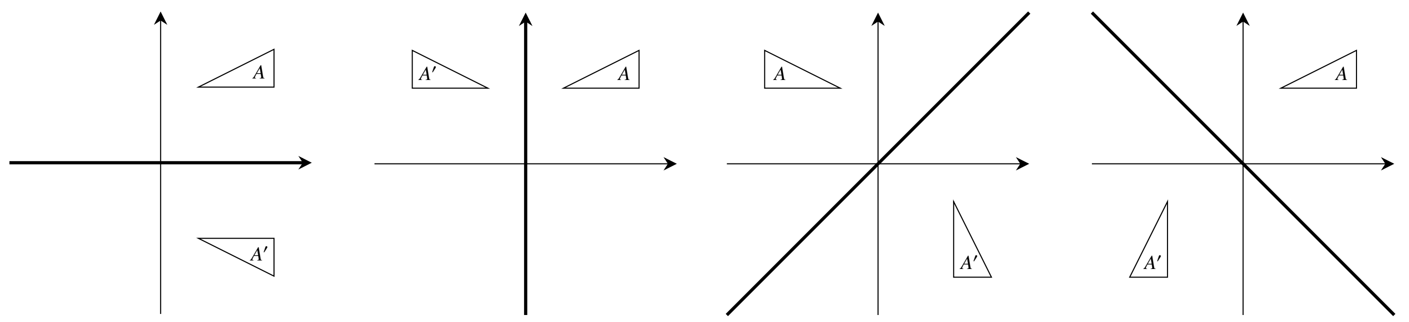
Reflection

Reflection is to reflect an object/shape about some line of symmetry/mirror line.
There are two cases under reflection:

Case 1

To be given an object/shape, $A(x, y)$ with a line of reflection, x-axis, y-axis, line $y = x$ or line $y = -x$, it asks for the Image of A. The method is:
E.g

$A(x, y) \mapsto A'(x, -y)$ $A(x, y) \mapsto A'(-x, y)$ $A(x, y) \mapsto A'(y, x)$ $A(x, y) \mapsto A'(-y, -x)$



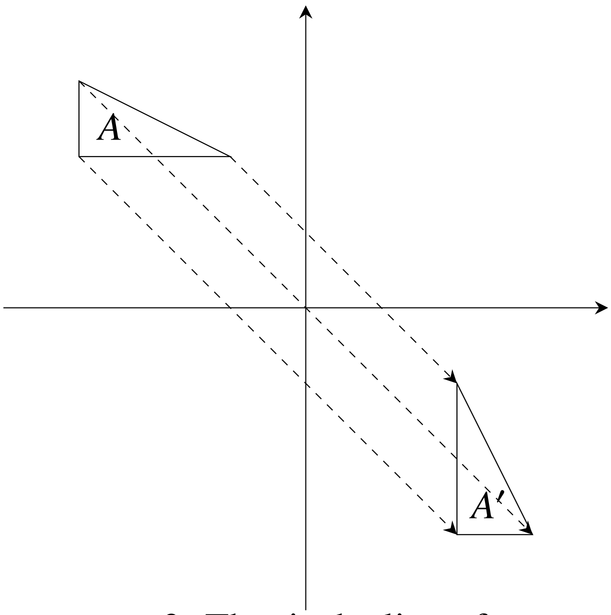
x-axis y-axis line $y = x$ line $y = -x$

Case 2

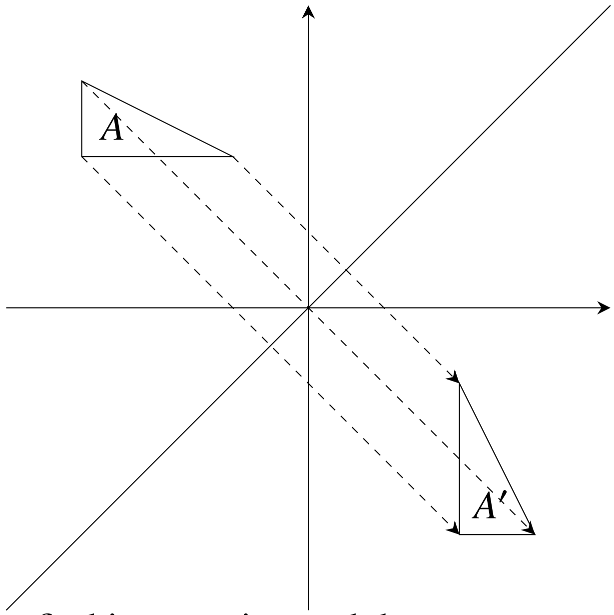
To be given an object/shape, $A(x, y)$ with its Image A', it asks for the line of reflection. The method is:

E.g

1. Join A to A'

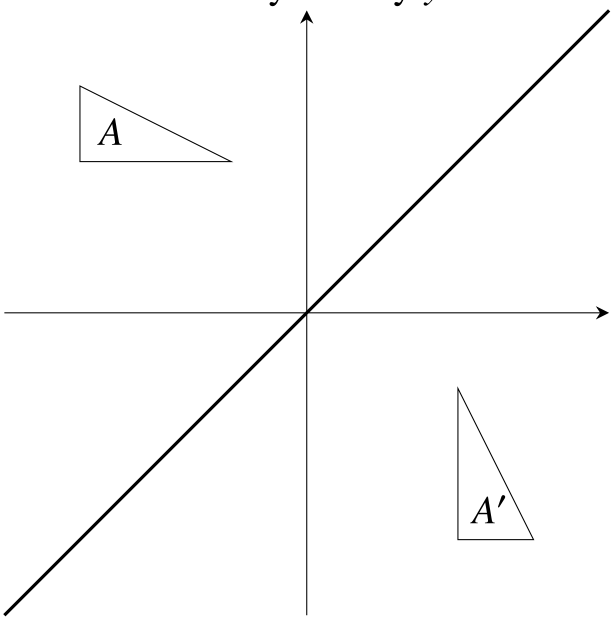


2. Draw their Perpendicular Bisectors.



3. That is the line of symmetry, find its equation and done.

$\therefore$ line of symmetry $y = x$



Rotation

Rotation is to rotate an object/shape about a point.
There are two cases under reflection.

Angles of Rotation given can be:
90° clockwise, which is = 270° anticlockwise
90° anticlockwise, which is = 270° clockwise
180°, which doesn't need a direction.

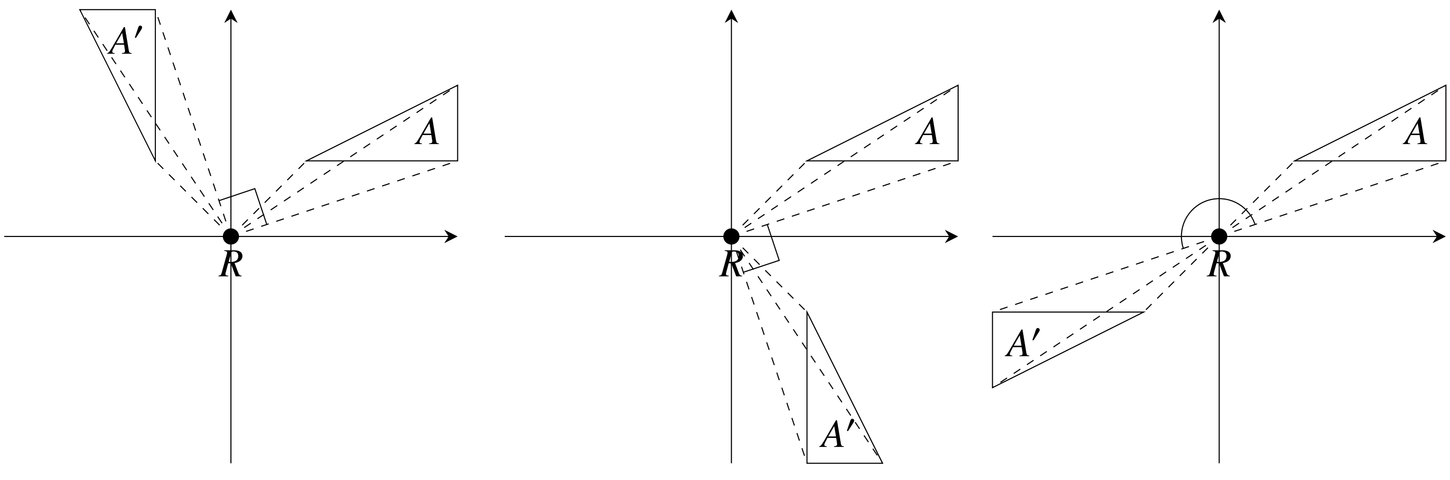
Case 1

To be given an object/shape, $A(x, y)$ with the centre of rotation R, with the angle and direction of rotation, it asks for the Image of A. The method is:
E.g
For cases with rotation of centre as $(0, 0)$.

Rotation of 90°, anticlockwise $A(x, y) \mapsto A'(-y, x)$

Rotation of 90°, clockwise $A(x, y) \mapsto A'(y, -x)$

Rotation of 180° $A(x, y) \mapsto A'(-x, -y)$

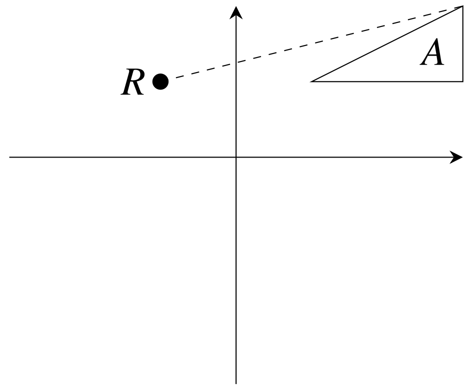


Rotation

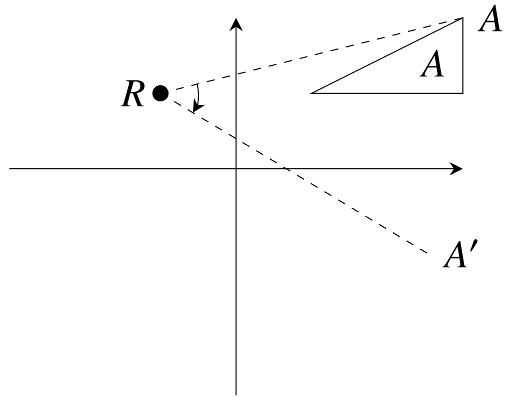
We can alternatively use a general method for centre of rotation and angles not being the above 3.

E.g for a given centre of rotation, $(-1, 1)$, and angle of 45° clockwise,

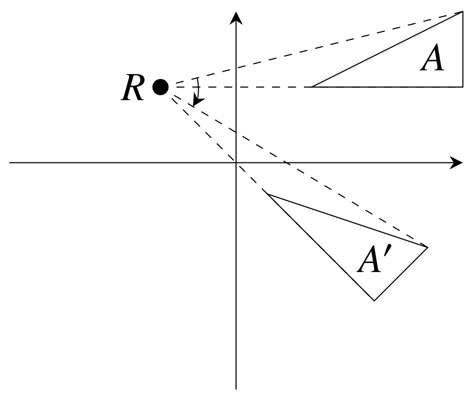
1. Draw line to the centre of rotation



2. Rotate accordingly to the question.
   Make the new line the same length as OA



3. Repeat for the other vertices
4. Complete the shape′

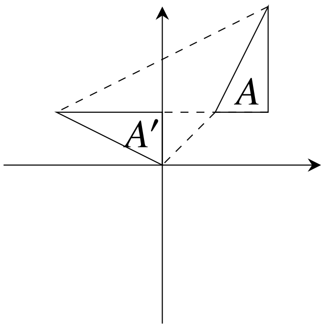


Rotation

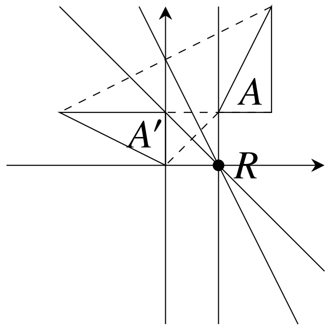
Case 2

To be given an object/shape, $A(x, y)$ with its Image A', it asks for the centre of rotation R, with the angle and direction of rotation. The method is:
E.g

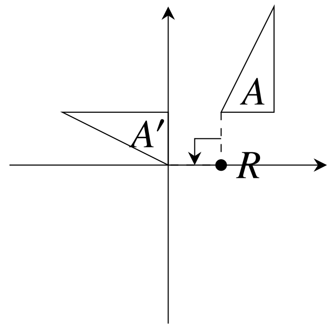
1. Connect vertices of A to vertices of A'



2. Draw their perpendiculars. Their intersection is the centre of rotation.



3. Join one A and A' pair to O. Measure angle. Direction is $A \to A'$.

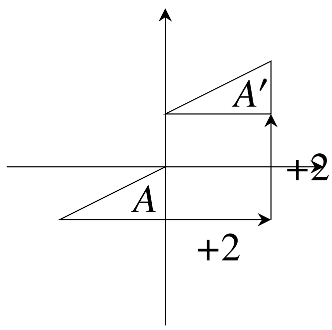


the answer we get is: ”90° anticlockwise rotation at center (1,0)”

Translation

Translation is to move an object/shape by some column vector.

Translation is given as $T = \begin{pmatrix} a \\ b \end{pmatrix}$. applying translation to A means $A(x, y) \to A'(x+a, y+b)$.
E.g



$$A + \begin{pmatrix} 2 \\ 2 \end{pmatrix} = A'$$

Enlargement

Engagement is to increase or decrease the size of a object/shape.

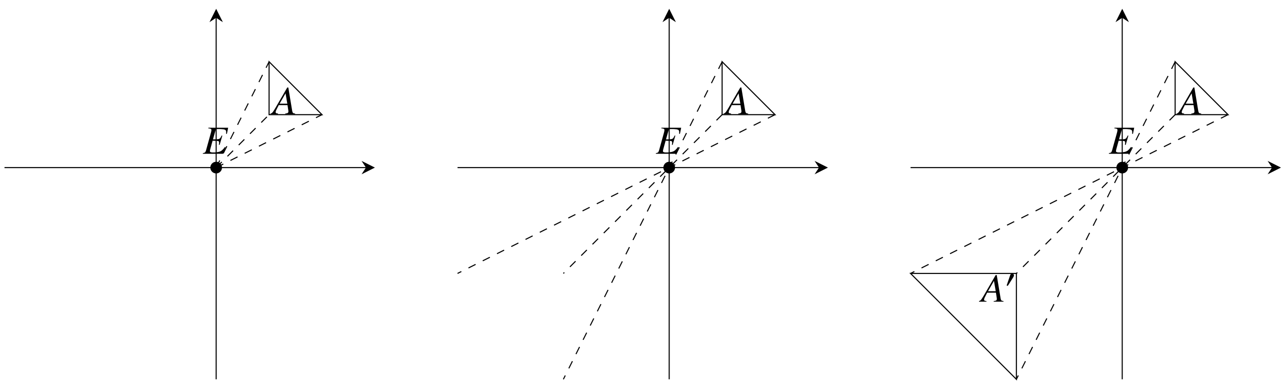
Scale factor, k, is found by. Either dividing 1 side of shape A' by A, e.g $k = \frac{A'B'}{AB}$. Or diving length of $A'E$ by AE, where E is the centre of enlargement, e.g $k = \frac{A'E}{AE}$

When points under go a enlargement, their distance from the centre of enlargement, E, is scaled by $|k|$, similarly each length of the of figure is $|k|$ times the original length.
If $|k| > 1$, the image is magnified/enlarged in size.
If $|k| < 1$, the image is reduced/shrunk in size.
If k is positive, the image lies on the same side/direction of centre of enlargement.
If k is negative, the image lies on the opposite side/direction of centre of enlargement and the image is inverted/flipped 180°.

Case 1

To be given an object/shape, $A(x, y)$, scale factor of k, and the centre of enlargement, E, it asks for the Image of A. The method is:
E.g for centre of enlargement, E, at $(0, 0)$ and scale factor of -2

1. Measure from centre to vertices of A
2. Apply scale factor k, remember the effects
3. Draw A'



Enlargement

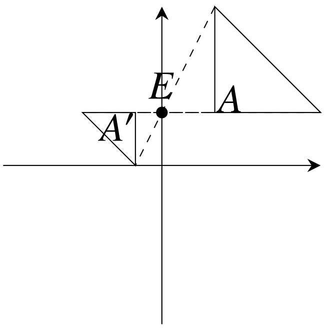
Case 2

To be given an object/shape $A(x, y)$, and its Image A', it asks for the centre of enlargement, E, and the scale factor k. The method is:
E.g

1. Join A to A' and extend the lines. Intersection is centre of enlargement.

2. If the image is on the same side, k is positive. If the image is on the opposite side, k is negative.

3. For value of k, its found by method told in page 116.



Opposite side,
aka $k < 0$

$k = \dfrac{A'B'}{AB} = \dfrac{1}{2}$
or
$k = \dfrac{A'E}{AE} = \dfrac{0.5}{1} = \dfrac{1}{2}$
Hence, $k = -\dfrac{1}{2}$

If it asks for the area of A', it will be $A' = k^2 \times A$

Repeated transformations

In this if we have, e.g $RT(A)$, then we perform $T(A)$ first then apply transformation R. A is the object, R and T are transformations having a variable assigned to them.

Inverse transformations

Inverse transformations are such they undo the action of the original transformation. e.g $T(A) = A'$, so $T^{-1}(A') = A$.
Rotation will flip its direction of rotation, from clockwise to anticlockwise and anticlockwise to clockwise.
Reflection will reflect the image again on the line of reflection.
Translation will move the object in the opposite direction. e.g $T = \begin{pmatrix} a \\ b \end{pmatrix}$, then $T' = \begin{pmatrix} -a \\ -b \end{pmatrix}$.

Vectors

Covering

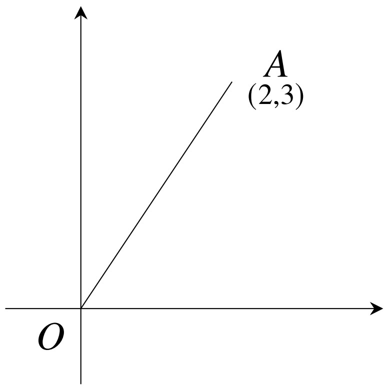
- Vectors in two dimensions — O levels and Extended
- Magnitude of a vector — O levels and Extended
- Vector geometry — O levels and Extended

Vectors A quantity which has a magnitude and a direction is called a vector. Length of the line is the magnitude and the tip ">" is the direction.

Vectors are represented in 2 ways. 1 is using $\overrightarrow{OA}$, or with a bold letter e.g **a**.

Position vectors describe the translation by vectors, e.g **i** or **j**. They are in column vector form $\begin{pmatrix} x \\ y \end{pmatrix}$.

E.g $\mathbf{a} = \overrightarrow{OA} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ is a Position Vector.



Position vectors are always from the origin, O.

Negative/Reverse vectors

A vector with the same magnitude but opposite direction. If $\overrightarrow{AB} = \begin{pmatrix} x \\ y \end{pmatrix}$ then the opposite/negative of $\overrightarrow{AB}$ is $\overrightarrow{BA} = \begin{pmatrix} -x \\ -y \end{pmatrix}$

Scaling Vectors

When vector is scaled by some constant k, it is written as $k\mathbf{a} = \begin{pmatrix} kx \\ ky \end{pmatrix}$

Parallel Vectors

If one vector is a multiple of another vector, then given vectors are called parallel vectors. If $\mathbf{a}$ and $\mathbf{b}$ are parallel vectors, they can be written as: $\mathbf{a} = k\mathbf{b}$, where k is any constant.

E.g If $\mathbf{p} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} k \\ 1 \end{pmatrix}$, and $\mathbf{p}$ and $\mathbf{q}$ are parallel, find k.

$$\text{Parallel;}$$
$$\mathbf{p}{=}h\mathbf{q}$$
$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = h \begin{pmatrix} k \\ 1 \end{pmatrix}$$
$$2 = hk$$
$$3 = h$$
$$\therefore 2 = 3k, \quad k = \frac{2}{3}$$

Equal vector

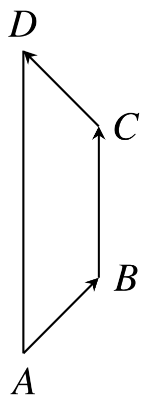
If two vectors have the same coordinates/translation, they are called equal vectors.

Addition/Subtraction of vectors

Adding a vector is interpreted as 2 ways:

Head to Tail

In this method, the head of one vector joins the tail of the next vector. The resultant vector of all vectors can be obtained by joining the tail of the first vector and head of the last vector. E.g we are given $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CD}$. If we wish to find $\overrightarrow{AD}$, we add up the vectors from head to tail: $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$. Next page's diagram shows it



This method can be used when all the vectors in its addition are given.
In head to tail, we cannot directly move from one point to another, i.e I can't do $A \rightarrow C$, but need to do $A \rightarrow B \rightarrow C$ in proper order, unless $A \rightarrow C$ is defined/given already.

Position vector

Position vector of a point, X, is the vector $\overrightarrow{OX}$, where O is the Origin, aka (0,0)
We can calculate the position vector of 2 points using the formula $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$, where $\overrightarrow{AB}$ is the position vector, $\overrightarrow{OA}$ is the position vector of the initial point and $\overrightarrow{OB}$ is the position vector of the final point.
E.g If $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, find position vector of A when $B = (1, 2)$

$$B = (1, 2), \ \therefore \ \overrightarrow{OB} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$
$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \overrightarrow{OA} - \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$
$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \overrightarrow{OA} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

Magnitude of vector

Stated by the absolute/modulus's 2 |s, we use Pythagoras. E.g $\left|\overrightarrow{OX}\right| = \left|\begin{pmatrix} x \\ y \end{pmatrix}\right| = \sqrt{x^2 + y^2}$

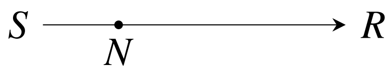
Ratio method

Ratio method is applied to find resultant vector when ratio of lines are given and one vector of given ratio is also given on that line.

They are given as either $OA : AB = a : b$ or as a equality $OA = \frac{a}{b}AB$, where OA and AB are parts of the full vector OB. This means, if given AB, $OA = \frac{a}{a+b}OB$ and $AB = \frac{b}{a+b}OB$.

In ratio method, we move directly from one point to the other point. To apply ratio method first assign ratios on parts of given line, then apply formula.

E.g N lies on SR such that $NR = 3SN$. Find SN



Here we see we can easily find SN as we got the ratio.

$$\frac{NR}{SN} = 3, \quad SN = \frac{NR}{3}$$

Facts about vectors and points

Parallel

We can state 2 vectors are parallel by showing one vector is a multiple of the second vector. Then we just write e.g "**a** and **b** are parallel because $\mathbf{a} = k\mathbf{b}$"

E.g $\mathbf{a} = 3\mathbf{i} + \mathbf{j}$, "**a** and **b** are parallel"
$\mathbf{b} = 9\mathbf{i} + 3\mathbf{j}$

Scalar

If we show $a = k\mathbf{b}$, we can state "**b** is k times **a**"
E.g if $\mathbf{a} = 2\mathbf{b}$, "**b** is twice of **a**"

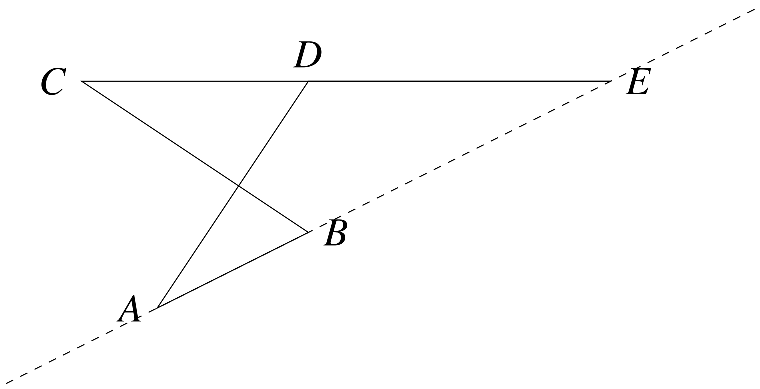
Equal

We can state 2 vectors are equal by solving for both and showing the 2 vectors have the same position vector (equation).

Facts about vectors and points

Collinear/Lie on a straight line

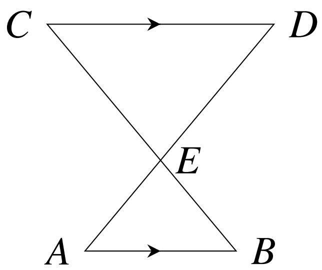
We can say 2 or more points are collinear if they: 1. are scalar multiple of each other AND 2. share the same point. E.g Here A, B, and E are collinear because $AB = k_1AE = k_2BE$ and that AB and BE share the point E.

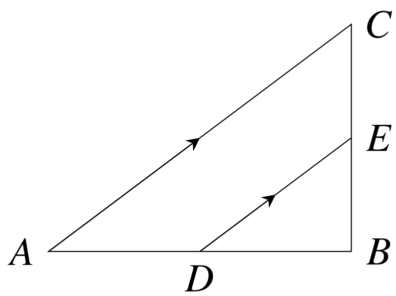


Area in Vectors

If the 2 shapes do not have a common height, but are similar. We use $\dfrac{A1}{A2} = \left(\dfrac{l1}{l2}\right)^2$

There are 2 cases. Both triangles are similar





In the first, opposite sides are parallel, e.g AB and CD, and l_1 and l_2 are on the same line, e.g AD and BD

In the second, the third sides are parallel, e.g AC and DE, and pairs of sides lie on the same line, e.g AB and DB or BC and EC.

If the shapes however, have the same height or have two vectors parallel, then we use $\dfrac{A1}{A2} = \dfrac{l1}{l2}$ where l_1 and l_2 are any 2 corresponding sides, or parallel vectors